

FIRST. / LAST.

Make row order part of the request

01 / THE SAS EXAMPLE

```
data test;
  input id group $ value;
  datalines;
1 A 10
2 A 20
3 B 30
4 B 40
5 B 50
6 C 60
;
run;
data flagged;
  set test;
  by group;
  first_flag = first.group;
  last_flag = last.group;
run;
proc print data=flagged noobs;
run;
```

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

State the ordering assumptions that define a row-based result.

These six records are sorted by group. FIRST.group and LAST.group mark each group's beginning and end. Group C has one member, so its row receives both flags.

EXPECTED RESULT

id	group	value	first_flag	last_flag
1	A	10	1	0
2	A	20	0	1
3	B	30	1	0
4	B	40	0	0
5	B	50	0	1
6	C	60	1	1

Scope: one nonmissing grouping variable and the supplied sorted rows. Both flags are 1 for a one-row group. All six rows remain in the output.

Expected values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

```
Convert this SAS code to Python.  
Show code in this response. Do not create or run files, or read  
existing Python files.
```

LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

TRY IT / THREE FRESH ATTEMPTS

Run A three times with the same SAS example and setup. Save each first response before asking follow-up questions. Record agreement as well as differences.

Bedrock A/B trials for this example are pending.

WHY CHOICES REMAIN OPEN

1 The source carries clues

The ordered records and BY statement may be enough for a correct translation. Observe the actual response.

2 The method is open

Adjacent-row comparisons or grouping methods can both work here. Different code can produce the same flags.

3 Ordering is implicit

The request does not explicitly preserve every row and its order. Those conditions deserve a direct statement.

ANTI-PATTERN TO AVOID

Anti-pattern: "Keep the first and last group records." That requests filtering. This code flags every row. Say which operation you need and verify the six-row output.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

Convert this SAS code to Python using pandas.
Show code in this response. Do not create or run files, or read existing Python files.

- Recreate id, group, and value. Keep all six rows in their order.
- These rows are already sorted by group, with no missing groups.
- Add first_flag and last_flag as integer 0/1 columns.
- first_flag is 1 on the first row or when group differs from the previous row. Compare with group.shift(1).
- last_flag is 1 on the last row or when group differs from the next row. Compare with group.shift(-1).
- Handle the first and last rows explicitly. Do not filter rows or reorder records within groups.
- Include assertions for id = [1, 2, 3, 4, 5, 6], first_flag = [1, 0, 1, 0, 0, 1], and last_flag = [0, 1, 0, 0, 1, 1].
- Return one complete script. Compute the flags from group; do not hard-code them. Explain why row 6 has both flags.

WHY THESE DETAILS HELP

- 1 Preserve the line**
Keeping row order protects the meaning of first and last. The sorted-input assumption is visible.
- 2 Name the operation**
Requesting flags preserves all records. A group summary would not satisfy the request.
- 3 Check the single-member group**
Group C checks that both flags can be true. Complete lists also check the interior rows.

CHECK IT YOURSELF

Run the script separately. Compare both flag lists and the id sequence. Row 6 must have both flags. A correct row count alone does not prove correct order or flags.

When first, last, or previous matters, state the row order and whether to keep every row.